Hoang Khuong **Duy**

AI RESEARCHER

FPT University, Ho Chi Minh City, Vietnam

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"Learning to solve real-world problems with AI"

Summary

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Recent Artificial Intelligence graduate with research experience in computer vision, representation learning, and robust visual understanding under distribution shift. Experienced in developing deep learning systems for medical imaging and industrial anomaly detection, with publications on long-tailed chest X-ray classification, device-invariant learning, and reconstruction-based anomaly detection. Skilled in PyTorch, TensorFlow, OpenCV, Docker, AWS, and reproducible experiment tracking. Interested in data-centric AI, generative modeling, and foundation-model-based representation learning for reliable real-world vision systems.

Education

VinUniversity & Vingroup — *Hanoi, Vietnam*

AI TALENT TRAINING PROGRAM, BASIC TRACK — *May 28, 2026 – Current*

- Participating in a practical AI talent training program implemented by VinUniversity in collaboration with Vingroup.
- Training includes foundation modules on AI thinking, AI ethics, AI skills, and problem solving with AI assistants, followed by hands-on work on real AI projects with enterprise mentors.

FPT University — *Ho Chi Minh City, Vietnam*

B.S. IN ARTIFICIAL INTELLIGENCE — *2023 - 2026*

- GPA: 3.4/4.00
- Thesis/Capstone: *CXR-DR: Chest X-ray Images Diagnosis and Report Generation*. Proposed a device-invariant, long-tailed chest X-ray diagnosis framework combining specialized ensemble CNNs, attention-guided feature learning, adversarial scanner/domain debiasing, and PubMedCLIP-based input validation for rare-pathology detection and cross-dataset robustness. Document | Code
- Relevant coursework: Deep Learning, Computer Vision, Machine Learning, Data Mining
- Merit 70% Scholarship, FPT University

Work Experience

AICLab, FPT University — *Ho Chi Minh City, Vietnam*

RESEARCH ASSISTANT — *12/2024 – Current*

- Working under the supervision of Dr. Huynh Cong Viet Ngu.
- Conducted research on robust visual representation learning for real-world distribution shifts, with applications in medical imaging and industrial anomaly detection.
- Developed novel learning pipelines for long-tailed classification, device-invariant representation learning, and reconstruction-based anomaly detection.
- Contributed to problem formulation, experimental design, ablation analysis, and manuscript preparation for peer-reviewed submissions.

FPT Software — *Ho Chi Minh City, Vietnam*

BACKEND INTERN — *6/2024 – 10/2024*

- Worked on Carus, a web-based automotive sales management platform.
- Contributed to backend service development in an industry software team, translating product requirements into maintainable APIs and data workflows.
- Implemented modular backend components with attention to reliability, traceability, and collaboration in a production-oriented codebase.

Publications

* denotes equal contribution.

Published and Accepted Conference Papers

1. Tri Nguyen*, **Duy Hoang***, and Ngu Huynh, "Statistical Non-linear Reconstruction Loss for Image Anomaly Detection," in *KES*, 2026. Accepted. (Core B).
   - Proposed a statistical non-linear reconstruction loss for image anomaly detection.
2. **Duy, H.K.**, et al., "Mitigating Class Imbalance in Chest X-Ray Classification With Memory-Augmented Models," in *ISCIT*, 2025. (Core B).
   - Proposed a memory bank module to retain stable pathological patterns and improve multi-label chest X-ray classification under class imbalance.
3. **Duy, H.K.**, et al., "Momentum-Anchored Multi-Scale Fusion for Long-Tailed Chest X-Ray Classification," in *FETC*, 2025. **Best Paper Award**.
   - Introduced EMA-based momentum anchoring with multi-scale fusion to stabilize representations for long-tailed chest X-ray classification.
4. Duy, N.H., **Duy, H.K.**, and Ngu, H.V.C., "Improving Imbalanced Multi-Label Chest X-Ray Diagnosis via CBAM-Enhanced CNN Backbones," in *FETC*, 2025.
   - Integrated CBAM into CNN backbones for imbalanced multi-label thoracic disease diagnosis. Code: GitHub.

Preprints

1. **Duy, H.K.**, Tri, N.M., and Ngu, H.C.V., "BoRAD: Bootstrap your Own Representations for Multi-class Anomaly Detection." *arXiv preprint arXiv:2606.14129*, 2026.
   - Proposed BoRAD, a label-free training framework for one-for-all multi-class anomaly detection. Preprint: arXiv.

Manuscripts Under Review

1. Nguyen, T.K., **Duy, H.K.**, and Ngu, H.V.C., "Device-Invariant Chest X-ray Classification via Attention and Adversarial Learning." Under review.
   - Combined self-attention with domain-adversarial learning to reduce device-related confounding in chest X-ray diagnosis. Code: GitHub.
2. Tri, N.M.*, **Duy, H.K.***, and Ngu, H.C.V., "Dual-Memory Based Reconstruction Method for Image Anomaly Detection." Under review.
   - Introduced DuMeNet, a dual-memory reconstruction network for image anomaly detection. Code: GitHub.

Honors & Awards

Research Awards

2025	**Best Paper Award**, FETC 2025	
2025	**First Runner-Up**, National FPT Student Research Festival	*Vietnam*
2025	**First Prize**, FPT HCMC Student Research Festival, Fall Semester	*Ho Chi Minh City, Vietnam*
2025	**Second Prize**, FPT HCMC Student Research Festival, Summer Semester	*Ho Chi Minh City, Vietnam*
2024	**First Prize**, FPT HCMC Student Research Festival, Fall Semester	*Ho Chi Minh City, Vietnam*

Academic Honors & Scholarships

Summer 2025	**Top 100 Outstanding Student**, FPT University	*Ho Chi Minh City, Vietnam*
2023–2026	**Merit 70% Scholarship**, FPT University	*Ho Chi Minh City, Vietnam*

Skills

Programming	Python, SQL, LaTeX
Deep Learning	PyTorch, TensorFlow, Keras, CNNs, attention, representation learning
Computer Vision	Medical imaging, anomaly detection, image reconstruction, OpenCV
Research Tools	Experiment tracking, ablation studies, reproducible pipelines, Git
Engineering	Docker, AWS, REST APIs, backend development
Languages	Vietnamese, English (IELTS 6.5)